

Francisco Portillo López

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LinkedIn | Personal website

RESEARCH INTERESTS

Cognitive plausibility of language and speech models; computational psycholinguistics; language acquisition; natural language processing.

EDUCATION

2023 – 2027 **B.A. in Spanish Language and Literature**
Universidad de Navarra, Pamplona, Spain
Undergraduate Thesis (TFG) in progress: *Cognition and Computation: Parallels and Divergences in Lexical and Computational Prediction*
Advisor: Prof. Mark Gibson

RESEARCH EXPERIENCE

Sep 2026 – present **Research Assistant | Language, Learning & Cognition Lab**
University of Louisville (*Remote*)
Supervisor: Dr. Layla Unger

- Developed naturalistic narrative stimuli for experimental research on forward and backward mechanisms in language processing.
- Designed and refined adult-oriented narratives according to controlled linguistic and experimental criteria, ensuring naturalness, coherence, and experimental validity.

June 2026 – present **Research Assistant | PhonLab**
Harvard University (*Remote*)
Supervisor: Dr. Kathryn Franich

- Supporting experimental design and quantitative analysis of speech-gesture coordination.
- Annotating and analyzing speech-gesture temporal alignment in ELAN.

- March 2026 – present **Research Assistant | Language and Computation in Neural Systems Group**
 Max Planck Institute for Psycholinguistics (*Remote*)
 Supervisor: Dr. Noémie te Rietmolen
- Extracted and validated prosodic features (F0, energy, duration) from French and Dutch MEG stimuli using the Helsinki wavelet toolbox.
 - Modeled MEG responses via mutual information and TRF analysis to test whether prosody improves neural encoding over an acoustic/lexical baseline.
- Jun 2026 – Aug 2026 **Research Intern | Infant Language & Cognition Group**
 Basque Center on Cognition, Brain and Language (BCBL), San Sebastián, Spain
 Supervisor: Dr. Olivera Savic
- Designed and ran an online cloze norming study on Spanish pseudowords, from participant recruitment through data cleaning.
 - Built R pipelines comparing human cloze probability, entropy, and surprisal against GPT-2 surprisal to test the cognitive plausibility of language models.
- March 2026 – August 2026 **Research Assistant | Clinical Linguistics & Disparities Lab**
 The University of Iowa (*Remote*)
 Supervisor: Dr. Philip Combiths
- Analyzing child-language corpora from populations with language disorders using Phon.
 - Applying acoustic and phonetic analysis to study linguistic variability and impairment.
- Sep 2025 – Jun 2026 **Research Assistant | Institute for Culture & Society (ICS)**
 Universidad de Navarra, Pamplona, Spain
 Supervisors: Dr. Inés Olza, Javier Yániz, Nadia Aldunate
- Conducting multimodal discourse analysis (ELAN, Praat) of interactional communication patterns.
- Jan 2025 – May 2025 **Research Assistant | Speech Laboratory**
 Universidad de Navarra, Pamplona, Spain
 Supervisor: Dr. Mark Gibson
- Supported Catalan and French phonetics projects, including Praat-based speech analysis.

PUBLICATIONS AND MANUSCRIPTS

Under Review	[3] Portillo, F. (August, 2026). <i>From Exposure to Expectation: Frequency, Surprisal, and Language Across Development in Spanish</i> . Manuscript submitted for publication. [2] Portillo, F. (January, 2026). <i>Artificial Rigidities vs. Biological Noise: A Comparative Analysis of Multisensory Integration in AV-HuBERT and Human Observers</i> . Available in SSRN: 10.2139/ssrn.6844390
Preprints	[1] Portillo, F. (November, 2025). <i>Imperfect Language, Artificial Intelligence, and the Human Mind: An Interdisciplinary Approach to Linguistic Errors in Native Spanish Speakers</i> . Preprint available in arXiv: 2511.01615

CONFERENCE PRESENTATIONS

June 2026	[4] <i>Normative Hallucinations: AI's Inability to Process the Cognitive Architecture of Spanish in the U.S.</i> Presented at the XI Symposium of the Instituto Cervantes Observatory at Harvard.
April 2026	[3] <i>Lenguaje imperfecto, inteligencia artificial y la mente humana: un enfoque interdisciplinar a los errores lingüísticos en hablantes nativos de español</i> . Presented at the XXIII Jornada Internacional de la Delegación de Venezuela ante la ALFAL.
March 2026	[2] <i>Imperfect Language, Artificial Intelligence, and the Human Mind: An Interdisciplinary Approach to Linguistic Errors in Native Spanish Speakers</i> . Presented at MHALC 2026, Maryland Hispanic Applied Linguistics.
December 2025	[1] <i>De reyes y carcajadas: parodia y burla en La infanta Palancona</i> . Presented at the XV Congreso Internacional Jóvenes Investigadores Siglo de Oro (JISO 2025).

HONORS, AWARDS & FELLOWSHIPS

2023 – 2027	Alumni Scholarship, Universidad de Navarra
February 2025	Winner, University Innovation Hackathon (top 10 projects nationally), Universidad de Navarra, Spain
June 2024	Second Prize, Certamen de Relato Corto de San Veremundo (Spain). Selected among over 100 participants

TECHNICAL SKILLS & METHODS

Computational	Python, R, Praat, ELAN, Phon
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Methods	TRF/mTRF analysis, corpus analysis (CHILDES, Wordbank), psycholinguistic experiment design, LLM surprisal & entropy modeling
Languages	Spanish (native), English (fluent), Latin (reading)

TEACHING EXPERIENCE

Sep 2025 – present	Writing Tutor, Writing Center Universidad de Navarra, Pamplona, Spain – Provided one-on-one writing tutoring for university students. – Developed teaching materials and conducted analyses using ELAN.
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PROFESSIONAL ACTIVITIES

Mar 2026 – Jun 2026	Member, Child's Play, Learning and Developmental Lab (Reading Group), University of Delaware
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